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ACTIVITY - 5

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Given input [4371, 1323, 6173, 4199, 4344, 9679, 1989] and a hash function $h(x) = x \pmod{10}$ show the resulting

a) separate chain hash table

b) Open addressing hash table using linear

Probing

c) Open addressing hash table using quadratic

Probing

d) Open addressing hash table with second hash function

Given,

a) separate chaining

$$h(4371) = 4371 \% 10 = 1$$

$$h(1323) = 1323 \% 10 = 3$$

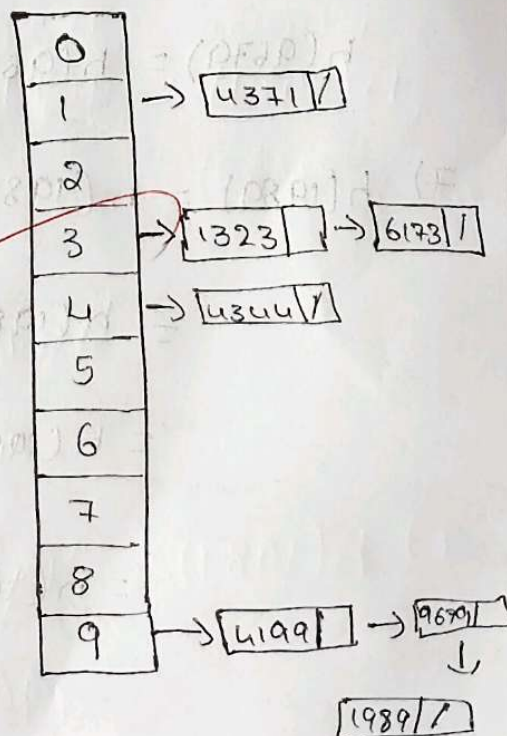
$$h(6173) = 6173 \% 10 = 3$$

$$h(4199) = 4199 \% 10 = 9$$

$$h(4344) = 4344 \% 10 = 4$$

$$h(9679) = 9679 \% 10 = 9$$

$$h(1989) = 1989 \% 10 = 9$$



b) Linear hashing

$$h(\text{key}) = h(\text{key} + i)$$

$$1) h(4371) = 4371 \% 10 = 1$$

$$2) h(1323) = 1323 \% 10 = 3$$

$$3) h(6173) = 6173 \% 10 = 3 \Rightarrow \text{collision}$$

$$h(6173) = h(6173 + 0) = h(6173) \% 10 = 3$$

collision
at $P=1$

$$h(6173) = h(6173 + 1) = h(6174) \% 10 = 4$$

$$4) h(4199) = h(4199) \% 10 = 9$$

$$5) h(4344) = h(4344) = 4 \Rightarrow \text{collision}$$

$$h(4344) = h(4344 + 0) = h(4344) \% 10 = 4 \Rightarrow P=1$$

collision

$$h(4344) = h(4344 + 1) = h(4345) \% 10 = 5$$

$$6) h(9679) = h(9679) \% 10 = 9$$

collision

$$h(9679) = h(9679 + 1) \% 10 = h(9780) \% 10 = 0$$

$P=2$

$$7) h(1989) = h(1989) \% 10 = 9$$

collision

$$= h(1989 + 1) \% 10 = h(1990) \% 10 = 0$$

$P=2$

$$= h(1989 + 2) \% 10 = h(1991) \% 10 = 1$$

collision

$$= h(1989 + 3) \% 10 = h(1992) \% 10 = 2$$

$P=3$
collision

		↑ 1989		↑ 6173					
0	1	2	3	4	5	6	7	8	9
↓	↓	↓	↓	↓				↓	
9679	4371	1323	4344					4199	

c) quadratic probing

$$h(\text{key}) = h(\text{key} + i^2) \Rightarrow i = 1, 2, 3, \dots$$

$$1) h(4371) = 4371 \% 10 = 1 \checkmark$$

$$2) h(1323) = 1323 \% 10 = 3 \checkmark$$

$$3) h(6173) = 6173 \% 10 = 3 \text{ collision}$$

$$h(6173) = h(6173 + 1) = h(6174)$$

$$h(6174) \% 10 = 4 \checkmark$$

$$4) h(4199) = 4199 \% 10 = 9 \checkmark$$

$$5) h(4344) = 4344 \% 10 = 4 \text{ collision}$$

$$h(4344) = h(4344 + 1^2) \% 10 = 5 \checkmark$$

$$6) h(9679) = 9679 \% 10 = 9 \text{ collision}$$

$$h(9679) = h(9679 + 1^2) \% 10$$

$$= h(9780) \% 10 = 0 \checkmark$$

$$7) h(1989) = 1989 \% 10 = 9 \text{ collision}$$

$$= h(1989 + 1) = h(1990) \% 10 = 0$$

collision

$$h(1989 + 2^2) = h(1993) \% 10 = 3 \text{ collision}$$

$$h(1989 + 3^2) = h(1998) \% 10 = 8 \checkmark$$

0	→ 9679
1	→ 4371
2	
3	→ 1323
4	→ 6174
5	→ 4344
6	
7	
8	→ 1989
9	→ 4199

d) Double hashing

$$h(\text{key}) = h_1(\text{key}) + (h_2(\text{key}) \times i) \bmod \text{size}$$

$$h_2(\text{key}) = R - (\text{key} \bmod R)$$

$$h_1(\text{key}) = x \bmod (T.S)$$

1) $h(4371) = 4371 \% 10 = 1$

2) $h(1323) = 1323 \% 10 = 3$

3) $h(6173) = 6173 \% 10 = 3$ collision

4) $h(6173) = 3 + [7 - 6] \times 1$
 $= 3 + 1 = 4 \checkmark$

5) $h(4199) = 4199 \% 10 = 9 \checkmark$

6) $h(4344) = 4344 \% 10 = 4$ collision

$h(4344) = 4 + [7 - 4] \times 1$
 $= 4 + 3 = 7 \checkmark$

7) $h(9679) = 9679 \% 10 = 9$ collision

$= 9 + (7 - 5) \times 1$

$= 11 \% 10 = 1$ collision

$= 9 + (7 - 5) \times 2$

$= (9 + 4) \% 10$

$= 13 \% 10 = 3$ collision

0	
1	→ 4371
2	
3	→ 1323
4	→ 6173
5	→ 9679
6	
7	→ 4344
8	
9	→ 4199

$$= 9 + (7-5) \times 3$$

$$= 9 + 6 = 15 \times 10 = 5 \checkmark$$

$$2) h(1989) = 1989 \times 10 = 9 \text{ collision}$$

$$= 1989 \times 10 + (7 - 1989 \times 10) \times 1$$

$$= 9 + (7-9) \times 1$$

$$= 9 + 6 = 15 \times 10 = 5 \text{ collision}$$

$$= 9 + (7-1) \times 2$$

$$= 9 + 12 = 21 \times 10 = 1 \text{ collision}$$

$$= 9 + (7-1) \times 3$$

$$= 9 + 18 = 27 \times 10 = 7 \text{ collision}$$

$$= 9 + (7-1) \times 4$$

$$= 9 + 24 = 33 \times 10 = 3 \text{ collision}$$

$$= 9 + (7-1) \times 5$$

$$= 9 + 30 = 39 \times 10 = 9 \text{ collision}$$

$$= 9 + (7-1) \times 6$$

$$= 9 + 36 = 45 \times 10 = 9 \text{ collision}$$

$$= 9 + (7-1) \times 7$$

$$= 51 \times 10 = 1 \text{ collision}$$

$$= 9 + (7-1) \times 8$$

$$= 57 \times 10 = 7 \text{ collision}$$

=

Good

Amal
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Note:-

$\therefore$ 1989 hash value doesn't have index position because i value has extend the tabular value